

Generative Engine Optimization (GEO) and Zero-Click Search in Sri Lanka

A Strategic Framework for Brand Visibility in the Age of Answer Engines

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Abstract

The diffusion of large language model (LLM)-based “generative engines” such as ChatGPT, Google AI Overviews, Gemini and Perplexity is reconfiguring the economics of online discovery. As answer engines synthesise responses directly within the results surface, the click — the foundational unit of the search-to-website transaction — is being decoupled from visibility. This paper examines the implications of this structural break for Sri Lankan brands, small and medium-sized enterprises (SMEs) and marketers, and advances Generative Engine Optimization (GEO) as the successor discipline to search engine optimization (SEO). Drawing on the foundational GEO framework of Aggarwal et al. (2024), contemporary global zero-click evidence (SparkToro, 2026; Bain & Company, 2025; Gartner, 2024; Pew Research Center, 2025), and Sri Lankan digital adoption data (DataReportal, 2025), the study makes three contributions. First, it consolidates the conceptual distinction between ranking-based and citation-based visibility, formalising the shift from “position one” to “source citation” as the new objective function of discovery. Second, it surfaces the specifically Sri Lankan dimensions of the transition — trilingual (Sinhala–Tamil–English) query behaviour, rural–urban connectivity asymmetry, low first-party data maturity, and SME skill constraints — that global GEO prescriptions do not address. Third, it proposes the GEO-LK Framework, a five-dimension readiness and intervention model (Citability, Structure, Trilingual Entity Authority, Statistical Evidence, and Measurement) tailored to resource-constrained, emerging-market contexts. The paper argues that Sri Lankan organisations face a narrowing window in which early GEO adoption can secure durable citation advantage before the category saturates, and that the cost of inaction is structural invisibility within the engines that increasingly mediate consumer and B2B decisions. Implications for practitioners, policymakers and marketing educators are discussed, and an agenda for empirical validation in the Sri Lankan setting is set out. The work is intended as a foundational reference for a domain that remains, at the time of writing, almost entirely unexamined in the Sri Lankan and broader South Asian scholarly literature.

Keywords: Generative Engine Optimization; GEO; Zero-Click Search; Answer Engine Optimization; Large Language Models; Digital Marketing; Brand Visibility; Sri Lanka; SMEs; Emerging Markets

Contents

1. Introduction

For three decades, the architecture of online discovery rested on a single, durable transaction: a user issued a query, a search engine returned a ranked list of hyperlinks, and the user clicked through to a website. The entire discipline of search engine optimization (SEO) was built to win that click, and an enormous global industry organised itself around the pursuit of “position one” on the results page. That transaction is now being dismantled. The arrival of large language model (LLM)-based systems — what Aggarwal et al. (2024) formalise as “generative engines” — has introduced a new paradigm in which the engine itself synthesises an answer from multiple sources and presents it directly to the user, frequently without any onward click to an external site. Google’s AI Overviews, Google AI Mode, OpenAI’s ChatGPT Search, Perplexity and Gemini are no longer experimental curiosities; they are the new front door to information.

The macro-evidence for this shift is now unambiguous. SparkToro’s longitudinal clickstream analysis finds that by early 2026, fewer than one in three Google searches in the United States still results in a click to the open web, with the zero-click rate reaching 68.01 per cent (SparkToro, 2026). Bain & Company (2025) report that eighty per cent of consumers now rely on AI-generated results for at least forty per cent of their searches. Gartner (2024) forecast that traditional search engine volume would fall by twenty-five per cent by 2026 as users migrate to conversational systems. ChatGPT alone reached roughly 900 million weekly active users by February 2026, more than doubling within a single year (OpenAI, 2026). The unit of value in discovery is no longer the rank; it is the citation — whether and how a brand is named, quoted and linked inside the synthesised answer.

This transformation gives rise to a successor discipline. Aggarwal et al. (2024) introduce the term Generative Engine Optimization (GEO) to describe “a novel paradigm to aid content creators in improving the visibility of their content” within generative engines, and demonstrate experimentally that targeted content interventions can raise a source’s visibility in AI-generated answers by as much as forty per cent. GEO is to the answer engine what SEO was to the link-list engine: the systematic practice of structuring, evidencing and positioning content so that it is selected, trusted and cited by the synthesising model. Where SEO optimised for crawlers and ranking algorithms, GEO optimises for retrieval-augmented generation and the citation behaviour of LLMs.

Yet the scholarly and practitioner conversation around GEO has been almost entirely Western and global in framing. The seminal contributions originate from North American computer science (Aggarwal et al., 2024; Chen et al., 2025), and the headline statistics describe United States and European behaviour. There is, at the time of writing, no substantive peer-reviewed treatment of GEO or the zero-click transition in the Sri Lankan context, and the South Asian literature more broadly remains focused on earlier-generation questions of social-media adoption and SEO. This is a consequential gap. Sri Lanka is a trilingual market in which discovery occurs across Sinhala, Tamil and English; it carries a pronounced rural–urban connectivity asymmetry; and its SME sector — the backbone of the economy — operates under acute

skill and budget constraints that make the global GEO playbook only partially transferable (DataReportal, 2025; Madurapperuma and Colombage, 2022).

This paper addresses that gap with three objectives. First, it consolidates the conceptual shift from ranking-based to citation-based visibility and positions GEO within the evolving landscape of search. Second, it situates the transition within Sri Lanka's specific digital reality, identifying the contextual factors that global prescriptions overlook. Third, it advances an original, named conceptual model — the GEO-LK Framework — intended both as a diagnostic instrument for Sri Lankan organisations and as a foundation for subsequent empirical work. The paper is deliberately written as a foundational reference: it aims to define the vocabulary, evidence base and research agenda for a domain that Sri Lankan scholarship has not yet entered.

2. From Search Engine Optimization to Generative Engine Optimization

2.1 The traditional search paradigm and its erosion

The classical search engine, as Aggarwal et al. (2024) observe, was powerful but fundamentally limited to returning a list of relevant websites in response to a query. Value accrued to whichever site occupied the highest organic position, because attention and clicks decayed steeply with rank. SEO emerged as the discipline of engineering that position through technical health, keyword relevance, backlink authority and content quality. For more than twenty years this model was stable, and the economics of digital marketing — including in Sri Lanka, where SEO and pay-per-click remain core agency offerings — were premised on its persistence (Madurapperuma and Colombage, 2022).

The erosion of that model did not begin with generative AI. It began with the gradual encroachment of “SERP features” — featured snippets, knowledge panels, People-Also-Ask boxes, local packs and inline calculators — each of which resolved a query without requiring a click. SparkToro's data shows the United States zero-click rate climbing from approximately fifty per cent in 2019 to 58.5 per cent in 2024 (SparkToro, 2026). What generative engines have done is convert this slow structural drift into a structural break. When an AI Overview is present, the zero-click rate rises to roughly eighty-three per cent, and organic click-through rates fall by as much as sixty per cent on those queries (Seer Interactive, 2025; Similarweb data reported in industry analyses, 2025). The results page has, in the words of one industry analyst, ceased to be a gateway and become a destination.

2.2 Defining the generative engine and the citation economy

A generative engine, following Aggarwal et al. (2024), is a system that satisfies a query by retrieving information from multiple sources and synthesising it into a single coherent response using an LLM. Architecturally it couples a retrieval mechanism (a search index or vector store) with a generative model that summarises, reconciles and attributes the retrieved material. The critical implication for marketers is that the engine, not the user, now performs the act of selection. The user receives a composed answer in which a small number of sources are named, quoted or linked, and the remainder are invisible. Discovery has shifted from a competition for rank to a competition for inclusion in the synthesis.

This gives rise to what may be termed the citation economy. In a zero-click environment, marketing value is created at the moment a brand is mentioned inside an AI-generated answer, irrespective of whether a click follows. The evidence indicates that such citation is far from cosmetic: brands cited within AI

Overviews earn an estimated thirty-five per cent more organic clicks and ninety-one per cent more paid clicks than uncited competitors on the same query (Seer Interactive, 2025), and AI-referred visitors convert markedly better and spend longer on-site than traditional organic visitors (Adobe, 2026; SE Ranking data reported in 2025). The strategic objective therefore migrates from ranking to being the source the model trusts enough to quote.

Table 1 formalises the distinction between the two paradigms across the dimensions most relevant to practitioners and researchers.

Table 1. Comparative anatomy of SEO and GEO.

Dimension	Search Engine Optimization (SEO)	Generative Engine Optimization (GEO)
Objective function	Achieve high rank on the results page	Be selected, trusted and cited inside the synthesised answer
Unit of success	The click	The citation / mention / quotation
Primary audience	Crawlers and ranking algorithms	Retrieval systems and the generative model's selection behaviour
Content target	Keyword-matched pages and backlinks	Quotable, evidence-rich, directly-answering passages
Winning content traits	Authority, keyword relevance, link equity	Citations, statistics, quotations, clear structure, fluency
Visibility surface	Ten blue links	AI Overviews, ChatGPT, Gemini, Perplexity, AI Mode
Measurement	Rank, organic traffic, CTR	Share of citation, prompt-level inclusion, AI-referred conversion
Failure mode	Page two of results	Structural invisibility (never cited)

Source: Author's synthesis, drawing on Aggarwal et al. (2024), SparkToro (2026) and Seer Interactive (2025).

2.3 The empirical foundations of GEO

The defining empirical contribution to date is the work of Aggarwal et al. (2024), presented at the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. The authors construct GEO-bench, a benchmark of thousands of queries spanning multiple domains, and test a battery of content interventions to determine which most reliably increase a source's visibility within generative-engine responses. Their central finding is that interventions which enrich the credibility and evidential density of content — adding authoritative citations, incorporating relevant quotations, and inserting pertinent statistics — are the most effective, raising visibility by up to forty per cent for well-chosen methods. Conversely, the keyword-stuffing tactics inherited from classical SEO are largely ineffective and can be counterproductive. A second wave of scholarship has begun to extend and stress-test these findings, examining domain-dependence, e-commerce settings, multi-query conflict, and the adversarial manipulability of conversational engines (Chen et al., 2025).

Table 2 summarises the principal optimisation levers identified in this emerging literature, with an indicative assessment of their effectiveness. The qualitative ratings synthesise the experimental directionality reported by Aggarwal et al. (2024) and subsequent work; precise magnitudes are domain-dependent and should be treated as directional rather than absolute.

Table 2. GEO content optimisation levers and indicative effectiveness.

Optimisation lever	Description	Indicative effectiveness
Citation addition	Attribute claims to credible, named sources	High
Quotation addition	Embed relevant expert or primary quotations	High
Statistics addition	Include precise quantitative evidence	High
Authoritative framing	Adopt a confident, expert register	Moderate
Fluency / readability	Improve clarity and linguistic quality	Moderate
Structured answering	Lead with a direct, concise answer; question-led headings	Moderate–High
Keyword stuffing	Repeat target keywords without added value	Negligible / negative

Source: Synthesised from Aggarwal et al. (2024) and Chen et al. (2025). Effectiveness is directional and domain-dependent.

2.4 A taxonomy of generative engines

The category “generative engine” is not monolithic, and effective optimisation requires distinguishing its principal forms, each of which exhibits different retrieval and citation behaviour. Three types are salient. The first is the embedded answer layer, exemplified by Google AI Overviews and AI Mode, in which generative summaries are grafted onto a conventional search index; here the legacy signals of SEO retain partial relevance because the underlying retrieval still draws on the web index. The second is the standalone conversational assistant, exemplified by ChatGPT and Gemini, which may answer from parametric memory, from a connected search tool, or from both, and whose citation behaviour is comparatively opaque. The third is the dedicated answer engine, exemplified by Perplexity, which is architected explicitly around retrieval and source attribution and therefore tends to cite more transparently. These types differ in how readily a brand can influence inclusion, in the visibility of their citations, and in the freshness of the content they surface.

For the Sri Lankan strategist, two consequences follow. First, GEO is not a single tactic but a portfolio calibrated to the engines a given audience actually uses; an export brand courting international B2B buyers faces a different engine mix from a domestic retailer serving a Tamil-speaking local market. Second, because the embedded answer layer still rests partly on web retrieval, classical SEO foundations — crawlability, authority, structured data — are not rendered worthless but are subsumed as necessary-but-insufficient conditions within the broader GEO discipline. GEO extends SEO; it does not simply replace it.

3. The Zero-Click Reality: Global Evidence

Any framework for the Sri Lankan context must be anchored in the global evidence base, both because that evidence establishes the magnitude of the shift and because Sri Lankan behaviour, with a lag, tends to track international platform dynamics (DataReportal, 2025). The data assembled in Table 3 is drawn from the most methodologically credible primary sources currently available, including clickstream-panel research, large-scale behavioural tracking and consumer surveys. Two cautions apply. First, zero-click figures vary by definition: some count any search without an external-website click, others count only sessions ending without any click whatsoever, and cross-year comparisons mix panel providers. Second,

several forecasts — notably the Gartner projection — have attracted scrutiny and appear to be unfolding unevenly across query categories. The figures are therefore best read as a converging body of directional evidence rather than as point-precise constants.

Table 3. Selected global evidence on zero-click search and AI-driven discovery.

Indicator	Value	Source
US zero-click rate (share of searches with no click)	~50%	SparkToro (2019)
US zero-click rate	58.5%	SparkToro / Datos (2024)
US zero-click rate	68.01%	SparkToro (2026)
Zero-click rate when an AI Overview is present	~83%	Similarweb data (2025)
Organic CTR decline on AI-Overview queries	up to ~60%	Seer Interactive (2025)
Forecast decline in traditional search volume by 2026	25%	Gartner (2024)
Consumers relying on AI results for $\geq 40\%$ of searches	80%	Bain & Company (2025)
ChatGPT weekly active users (Feb 2026)	~900m	OpenAI (2026)
Additional organic clicks for AI-Overview-cited brands	+35%	Seer Interactive (2025)
Additional paid clicks for AI-Overview-cited brands	+91%	Seer Interactive (2025)
B2B buyers using generative AI in the purchase process	94%	6sense (2025)
US marketers planning to implement GEO within 3–6 months	54%	eMarketer (2026)
Marketers currently investing in GEO measurement	23%	Incremys (2025)
YoY growth in AI referral traffic to US retail (2025 season)	+693%	Adobe (2026)
Share of LLM citations drawn from the first 30% of content	44.2%	SparkToro (2026)

Source: Compiled by the author from the cited primary and secondary sources. Definitions and panels differ across years; values are directional.

Three patterns in this evidence are strategically decisive. First, the decline in clicks is structural and accelerating rather than cyclical: the same panel that measured 58.5 per cent zero-click in 2024 measured 68.01 per cent by 2026 (SparkToro, 2026). Second, citation confers measurable advantage even in the absence of a click, which reframes the marketing objective around inclusion rather than traffic (Seer Interactive, 2025). Third, a significant adoption gap has opened between intention and capability: while a majority of marketers intend to act on GEO, only around a quarter currently measure it (eMarketer, 2026; Incremys, 2025). This gap is precisely the window of opportunity that early movers — including in under-served markets such as Sri Lanka — can exploit before the discipline matures and citation real estate becomes contested.

4. The Sri Lankan Digital Context

Sri Lanka enters the generative-engine era from a position of rapid but uneven digital maturation. According to DataReportal (2025), at the start of 2025 the country had 12.4 million internet users against

a population of 23.2 million, an online penetration of 53.6 per cent, alongside 8.20 million social-media user identities and 29.3 million registered mobile connections. Social-media identities grew by 9.3 per cent year-on-year, an addition of roughly 700,000 users. These headline figures, summarised in Table 4, describe a market that is digitally engaged at the level of social platforms but whose web-discovery behaviour is being reshaped by the same generative dynamics observed globally.

Table 4. Key digital indicators for Sri Lanka, January 2025.

Indicator	Value (Jan 2025)
Total population	23.2 million
Internet users	12.4 million
Internet penetration	53.6%
Social-media user identities	8.20 million (35.4% of population)
Year-on-year growth in social-media identities	+9.3% ($\approx +700,000$)
Registered mobile connections	29.3 million
Share of population that is rural	80.5%
Share of internet users using social media	66.0%
Facebook ad reach	8.20 million
YouTube ad reach	8.13 million
TikTok ad reach	5.79 million

Source: DataReportal (2025), *Digital 2025: Sri Lanka*.

Beneath these aggregates lie four contextual features that materially condition how GEO must be approached in Sri Lanka, and which distinguish the local problem from the global one.

Trilingual discovery. Sri Lankan information-seeking occurs across three languages — Sinhala, Tamil and English — frequently with code-switching within a single user’s behaviour. Generative engines vary substantially in their competence and citation behaviour across these languages, and Sinhala and Tamil are comparatively low-resource from the perspective of LLM training data. A brand that is well evidenced in English may be entirely absent from a Tamil-language synthesised answer. This trilingual fragmentation has no analogue in the predominantly English-language global GEO literature and is, in this author’s assessment, the single most distinctive feature of the Sri Lankan problem.

Rural–urban asymmetry. With 80.5 per cent of the population classified as rural (DataReportal, 2025) and national connectivity unevenly distributed, the audience reachable through generative engines is geographically skewed toward urban and peri-urban centres. Industry analyses place national internet access at 53.6 per cent but note that rural penetration lags substantially behind urban penetration. GEO strategies must therefore be paired with realistic audience forecasting and, for many SMEs, with continued investment in offline and social channels rather than treated as a wholesale replacement.

First-party data and skill maturity. The Sri Lankan SME sector is widely characterised by low digital-marketing maturity, ad hoc execution and limited analytics capability (Madurapperuma and Colombage, 2022). Where global GEO guidance assumes sophisticated content operations and

measurement infrastructure, the median Sri Lankan SME possesses neither. Any workable local framework must therefore be implementable with modest resources and must explicitly build measurement capacity rather than presuppose it.

An emerging local AI discourse. A nascent Sri Lankan literature on artificial intelligence in marketing and creative work has begun to appear, addressing employee attitudes toward AI in financial services (Dewasiri et al., 2023), human–AI collaboration in content creation (Wijenayake et al., 2025), and consumer acceptance of AI-generated advertising. This work signals scholarly appetite for the topic, but none of it addresses generative-engine visibility or the zero-click transition. The conceptual territory of GEO in Sri Lanka is, in short, open.

5. The GEO-LK Framework

To translate the global discipline of GEO into the Sri Lankan setting, this paper proposes the GEO-LK Framework: a five-dimension model that functions both as a diagnostic of an organisation’s readiness for the generative-engine era and as a prescriptive agenda for intervention. The framework integrates the empirically supported optimisation levers of Aggarwal et al. (2024) with the four contextual features identified in Section 4. Its five dimensions are Citability, Structure, Trilingual Entity Authority, Statistical and Evidentiary Depth, and Measurement — mnemonically, the C-S-T-S-M model.

Table 5. The GEO-LK Framework: dimensions, rationale and Sri Lankan application.

Dimension	Definition	Sri Lankan consideration	Example intervention
Citability	Make content the most quotable, source-worthy answer to a defined question	Most SME content is promotional, not citable; little primary evidence exists locally	Publish original, attributable claims and data engines can lift verbatim
Structure	Architect content for machine parsing and direct answering	Low schema and FAQ adoption among Sri Lankan SMEs	Question-led headings, concise lead answers, FAQ and structured data
Trilingual Entity Authority	Build a consistent, machine-recognisable entity across Sinhala, Tamil and English	Sinhala/Tamil are low-resource; entities fragment across languages	Parallel trilingual content; consistent naming; local-language citations
Statistical & Evidentiary Depth	Embed original statistics, named frameworks and primary research	Scarcity of local data makes original evidence disproportionately valuable	Commission and publish local surveys; cite Sri Lankan primary sources
Measurement	Track citation and share-of-voice inside answer engines, not rank alone	Near-absent GEO measurement capability locally	Prompt-level monitoring across ChatGPT, Gemini, Perplexity and AI Overviews

Source: Author’s original framework, June 2026.

5.1 Citability

The first dimension reframes the purpose of content. In the SEO paradigm, content existed to rank and to persuade the human who clicked through. In the GEO paradigm, the primary reader is the model, and the model preferentially selects content it can lift, attribute and trust. Citability is the property of being the most quotable answer to a specific question. The practical implication for Sri Lankan organisations —

whose digital content is overwhelmingly promotional and self-referential — is a difficult but high-leverage discipline: to publish claims that are precise, attributable and genuinely informative, even when they do not directly sell. Aggarwal et al. (2024) provide the empirical warrant: content enriched with citations, quotations and statistics is materially more likely to be surfaced. Citability is therefore not a stylistic preference but a measurable optimisation.

5.2 Structure

Generative engines reward content they can parse cleanly. Structured answering — leading with a concise, direct response to the implied question, organising material under question-led headings, and exposing machine-readable signals such as FAQ markup and structured data — increases the probability of selection. Industry evidence indicates that longer, conversational and question-form queries disproportionately trigger AI answers, which elevates the importance of long-tail, question-based content architecture. For the Sri Lankan SME, where schema adoption is rare, structure represents a low-cost, high-return intervention that requires discipline rather than budget.

5.3 Trilingual Entity Authority

This dimension is the framework’s principal original contribution and the feature most specific to Sri Lanka. An “entity” in this context is the machine-recognisable identity of a brand, person or organisation — the consistent cluster of attributes that an engine associates with a name. In a single-language market, building entity authority is a unitary task. In Sri Lanka it is threefold, because the engine’s knowledge of a brand in English does not transfer automatically to its knowledge in Sinhala or Tamil, both of which are lower-resource languages with thinner training corpora. A brand that has invested only in English content may be invisible to a Tamil-language query even where its English visibility is strong. Trilingual Entity Authority therefore calls for deliberate, parallel investment in Sinhala and Tamil content, consistent cross-language naming, and local-language citation, so that the entity is coherent and discoverable across all three linguistic surfaces of the market. This is simultaneously a barrier and an opportunity: it is costly, but precisely because few competitors will undertake it, early movers can secure durable citation advantage in the Sinhala and Tamil answer spaces.

5.4 Statistical and Evidentiary Depth

Because the most effective GEO levers involve statistics and citations (Aggarwal et al., 2024), and because credible local data is scarce in Sri Lanka, original evidence is disproportionately valuable here. An organisation that commissions and publishes even modest primary research — a local consumer survey, a sector benchmark, a named analytical framework — creates the very citable artefacts that engines preferentially surface, and does so in a domain where almost no competing evidence exists. This dimension reframes research and thought-leadership not as marketing overhead but as a direct visibility investment, and it aligns the commercial incentive of brands with the scholarly value of generating local knowledge.

5.5 Measurement

The final dimension confronts the adoption gap directly. Traditional analytics — rank tracking and organic-traffic reporting — are structurally blind to the citation economy, because they measure clicks that increasingly do not occur. GEO measurement instead tracks whether and how a brand is cited within answer engines: share of citation against named competitors, prompt-level inclusion across ChatGPT,

Gemini, Perplexity and AI Overviews, and the conversion quality of AI-referred visitors. Globally only around a quarter of marketers currently undertake such measurement (Incremys, 2025); in Sri Lanka the figure is almost certainly lower. Building this capability is the precondition for managing the other four dimensions, since visibility that cannot be measured cannot be optimised.

6. Illustrative Applications and Sectoral Implications

The GEO-LK Framework can be illustrated through the sectors in which it is most immediately consequential for Sri Lanka. These illustrations are analytical rather than empirical, and are offered to demonstrate the framework's explanatory reach pending the validation set out in Section 9.

Tourism and hospitality. Sri Lanka's tourism recovery depends on discovery by international travellers who increasingly plan through conversational engines. A query such as “best boutique hotels in the Sri Lankan hill country” is now frequently answered by synthesis rather than by a ranked list. Properties that are richly evidenced — with specific, citable detail, third-party corroboration and structured data — are candidates for inclusion; those whose web presence is purely promotional are not. Because the audience here is predominantly English-language and international, the Citability, Structure and Statistical-Depth dimensions dominate, and Trilingual Authority is secondary.

Apparel and export brands. For Sri Lanka's apparel and value-added export sector, generative engines mediate both B2B sourcing decisions and end-consumer discovery in overseas markets. With ninety-four per cent of B2B buyers reporting use of generative AI in their purchase process (6sense, 2025), an export brand's inclusion in AI-synthesised supplier comparisons is now a material commercial variable. Evidentiary depth — verifiable claims about capability, compliance and provenance — is the decisive lever.

Domestic SMEs and local discovery. For the domestic SME serving a Sinhala- or Tamil-speaking local market, Trilingual Entity Authority and local-intent structure are paramount. A consumer searching for a service “near me” in Sinhala expects a synthesised local answer; the SME that has invested in consistent local-language content and structured local signals is the one cited. Here the framework's most distinctive dimension is also the most valuable, and the competitive field is least contested.

Professional services and personal brands. For consultants, educators and founders building authority, the citation economy rewards the publication of named frameworks, original analysis and primary data — the artefacts most likely to be quoted by an engine. This is the mechanism by which an individual or firm becomes the source a model reaches for, and it aligns directly with the scholarly practice of producing citable, indexed work.

7. Risks, Manipulation and the Ethics of GEO

The migration to generative engines introduces risks that a responsible treatment of GEO must confront, and that are particularly acute in a market with limited regulatory infrastructure such as Sri Lanka. The first is hallucination. Generative engines synthesise rather than retrieve, and they are known to produce fluent but inaccurate statements, including misattributed or fabricated citations. A brand may find itself associated with claims it never made, or excluded in favour of a confidently stated error. Because the synthesised answer carries an authority the underlying sources may not warrant, the consumer's capacity to verify is diminished precisely as their reliance grows.

The second risk is adversarial manipulation. A growing strand of the literature demonstrates that the selection and ranking behaviour of conversational engines can be deliberately gamed — through crafted content, injected instructions and other techniques designed to inflate a source’s visibility irrespective of merit (Chen et al., 2025). This is the generative-era analogue of black-hat SEO, and it raises the prospect of a market in which citation is captured by the most manipulative rather than the most credible actor. For an emerging market, where digital literacy among both consumers and SME owners is uneven (Madurapperuma and Colombage, 2022), the asymmetry between sophisticated manipulators and unprotected users is a genuine consumer-protection concern.

These risks delineate an ethical boundary for the discipline. The GEO-LK Framework is deliberately constructed around legitimate, evidence-based visibility — citability earned through genuine informativeness, statistical depth grounded in real data, and entity authority built through consistent, truthful representation — rather than around manipulation. This is not merely a moral preference but a strategic judgement: engines and their operators are actively developing defences against manipulation, and visibility secured by gaming is inherently fragile, whereas visibility earned through credible evidence is durable. The ethical path and the effective path, in this analysis, converge. For policymakers and educators, the implication is that the response to the generative-engine era should pair the promotion of GEO capability with investment in digital literacy and, in time, with appropriate guidance on transparency and disclosure.

8. Methodological Note

This paper is conceptual and integrative in nature. Its evidentiary base comprises three strands: the peer-reviewed and pre-print computer-science literature that defines GEO (Aggarwal et al., 2024; Chen et al., 2025); a body of primary and secondary measurement of zero-click and AI-search behaviour from established research organisations (SparkToro, 2026; Bain & Company, 2025; Gartner, 2024; Pew Research Center, 2025; Seer Interactive, 2025; OpenAI, 2026; Adobe, 2026; 6sense, 2025); and authoritative descriptive data on the Sri Lankan digital environment (DataReportal, 2025), complemented by the emerging Sri Lankan literature on AI in marketing (Dewasiri et al., 2023; Wijenayake et al., 2025; Madurapperuma and Colombage, 2022). The GEO-LK Framework is derived analytically from the intersection of these strands. The framework’s propositions are, at this stage, theoretically grounded rather than empirically tested; the validation agenda is specified in Section 9. The author’s own practice as a performance-marketing practitioner informs the illustrative applications, consistent with the reflective practitioner stance adopted in prior work (Iyer Subramania Iyer, 2025).

9. Limitations and a Sri Lankan Research Agenda

Three limitations bound the present contribution. First, the global statistics on which the argument rests are predominantly United States and European in origin; Sri Lanka-specific zero-click and AI-citation measurement does not yet exist, and the assumption that local behaviour tracks global dynamics with a lag, while reasonable, is untested. Second, generative engines are evolving at a pace that will date specific figures quickly; the values reported here should be read as a snapshot of mid-2026. Third, the GEO-LK Framework is a conceptual instrument awaiting empirical validation.

These limitations define a concrete and, in the Sri Lankan setting, almost entirely unoccupied research agenda. Four lines of enquiry are proposed. (1) A direct measurement study of zero-click and AI-citation

rates for Sri Lankan queries across the three national languages, which would establish the first local baseline. (2) An empirical test of the GEO-LK Framework through a longitudinal intervention study with a cohort of Sri Lankan SMEs, measuring change in citation share following structured GEO investment. (3) A comparative analysis of generative-engine citation behaviour across Sinhala, Tamil and English, quantifying the trilingual visibility gap hypothesised in Section 5.3. (4) A policy-oriented study of the implications of the zero-click transition for Sri Lanka's SME competitiveness and for digital-marketing curricula, including at the institutions with which the author is associated. Each of these would constitute an original contribution to a literature that, at present, does not exist.

10. Conclusion

The mechanism of online discovery is undergoing its most fundamental change since the search engine itself. As generative engines synthesise answers and the click decouples from visibility, the strategic objective of digital marketing shifts from achieving rank to securing citation. Generative Engine Optimization is the discipline of that new objective, and the global evidence assembled here — a zero-click rate exceeding two-thirds, a measurable citation premium, and rapidly rising adoption of conversational engines — establishes that the shift is structural, not speculative.

For Sri Lanka, the transition arrives with distinctive contours: a trilingual discovery environment, a pronounced rural–urban asymmetry, a resource-constrained SME sector, and an almost complete absence of local scholarship or measurement. The GEO-LK Framework advanced in this paper — Citability, Structure, Trilingual Entity Authority, Statistical and Evidentiary Depth, and Measurement — is offered as the first systematic attempt to translate the global discipline into this setting, and as a foundation for the empirical work the field now requires. The central argument is one of timing. A gap has opened between the intention to act on GEO and the capability to do so, and that gap is widest in under-served markets. Sri Lankan organisations that build citation advantage now — particularly in the under-contested Sinhala and Tamil answer spaces — stand to secure a durable position before the discipline matures. The alternative is structural invisibility within the engines that increasingly decide what consumers and buyers come to know. The choice, for brands and for the scholarship that would guide them, is to enter the citation economy deliberately or to be omitted from it by default.

References

- 6sense (2025) *2025 Buyer Experience Report*. San Mateo, CA: 6sense Insights.
- Adobe (2026) *AI Traffic Surges Across Industries: Adobe Digital Insights*. San Jose, CA: Adobe Inc.
- Aggarwal, P., Murahari, V., Rajpurohit, T., Kalyan, A., Narasimhan, K. and Deshpande, A. (2024) 'GEO: Generative Engine Optimization', in *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '24)*. Barcelona: Association for Computing Machinery, pp. 5–16. doi:10.1145/3637528.3671900.
- Bain & Company (2025) *Goodbye Clicks, Hello AI: Zero-Click Search Redefines Marketing*. Boston, MA: Bain & Company.
- Chen, M., Wang, X., Chen, K. and Koudas, N. (2025) *Generative Engine Optimization: How to Dominate AI Search*. arXiv preprint arXiv:2509.08919.

- DataReportal (2025) *Digital 2025: Sri Lanka*. DataReportal / Kepios. Available at: <https://datareportal.com/reports/digital-2025-sri-lanka> (Accessed: 23 June 2026).
- Dewasiri, N.J., Pigera, A.K.M., Karunarathne, K.S.S.N. and Rathnasiri, M.S.H. (2023) 'Financial services employee engagement and attitude toward artificial intelligence: evidence from Sri Lanka', in *Transformation for Sustainable Business and Management Practices*. Bingley: Emerald Publishing, pp. 231–245.
- eMarketer (2026) *The State of Generative Engine Optimization Among US Marketers*. New York: eMarketer / Insider Intelligence.
- Gartner (2024) *Gartner Predicts Search Engine Volume Will Drop 25% by 2026*. Stamford, CT: Gartner, Inc.
- Incremys (2025) *GEO Measurement Adoption Benchmark*. Industry report.
- Iyer Subramania Iyer, V. (Sharma, S.) (2025) 'From Renovation to Innovation and Leading Change in the Creative Economy: a case study of organizational change, culture, and digital transformation at Prime One Global's Vavuniya office'. Working paper. ORCID: 0009-0008-1772-7114.
- Madurapperuma, R.K. and Colombage, V.K. (2022) 'Factors affecting the adoption of digital marketing by SMEs in the context of Sri Lanka', in *Proceedings of the International Conference on Business Innovation (ICOBI 2022)*. Homagama: NSBM Green University.
- OpenAI (2026) *ChatGPT Usage Update, February 2026*. San Francisco, CA: OpenAI.
- Pew Research Center (2025) *Google Users Are Less Likely to Click on Links When an AI Summary Appears*. Washington, DC: Pew Research Center.
- Seer Interactive (2025) *The Impact of AI Overviews on Organic and Paid Click-Through Rates*. Philadelphia, PA: Seer Interactive.
- SparkToro (2026) *In 2026, Less than One Third of Google Searches Still Send a Click*. SparkToro. Available at: <https://sparktoro.com> (Accessed: 23 June 2026).
- Wijenayake, S. et al. (2025) 'Generative AI in content creation: optimizing human–AI collaboration in the creative process of Sri Lanka'. Conference paper.